ASI COVID Manufacturing Recovery Analysis

Cross-Industry Evidence on Economic Shock, Structural Labour Vulnerability, and Rebound Archetypes in India

Course: ECO 6810 - Final Project	**Date:** Academic Year 2025–26
Team Members: Madhav Kumar, Vikas Chaurasiya	**Project Type:** Descriptive Empirical Analysis
Target Policy Body: DPIIT, Ministry of Commerce & Industry	**Status:** Approved Charter | Primary Threshold Passed

Executive Summary: This study investigates the heterogeneous economic impacts of the COVID-19 disruption and subsequent recovery across Indian manufacturing using factory-level survey microdata from the Annual Survey of Industries (ASI, MoSPI) covering 2019–20, 2020–21, and 2021–22. Aggregating data across 23 National Industrial Classification (NIC) 2-digit manufacturing sectors with official factory multipliers (MULT), we test whether labour-intensive industries experienced disproportionately larger Gross Value Added (GVA) losses during the 2020–21 shock year relative to capital-intensive sectors. We also deploy unsupervised K-Means clustering to identify recovery archetypes and Random Forest regression to explore pre-shock structural determinants of resilience.

Key Finding: Labour-intensive industries experienced an average COVID-era GVA contraction of **-4.71%**, compared to **-1.09%** in capital-intensive industries—an excess contraction of **3.63 percentage points**, successfully passing the pre-specified charter threshold (≥ 2.0 pp). The project satisfies all sample constraints with 23 eligible industries (each exceeding 500 survey factories per year).

1. Research Question & Institutional Stakeholder

The COVID-19 pandemic generated one of the most severe economic shocks to Indian registered manufacturing in modern history. However, aggregate macroeconomic contraction metrics mask immense structural disparities across specific industries. This project addresses three central empirical questions:

- **Shock Distribution:** Which manufacturing sectors suffered the largest Gross Value Added (GVA) contractions during the 2020–21 lockdown period?
- **Structural Vulnerability:** Did industries relying more heavily on labour relative to fixed capital experience systematically worse economic downturns?
- **Recovery Archetypes:** Which sectors rebounded strongest in 2021–22, and what distinct recovery patterns emerge across Indian manufacturing?

Policy Stakeholder: The primary institutional audience is the **Department for Promotion of Industry and Internal Trade (DPIIT)**, Ministry of Commerce and Industry, Government of India. DPIIT is responsible for industrial restructuring, emergency relief allocation, and long-term manufacturing growth. Understanding industry-specific vulnerability and recovery archetypes provides direct evidence for targeting relief packages and emergency capital buffers toward vulnerable sectors during future shocks, rather than relying on uniform, blanket subsidies.

2. Data Construction & Sampling Pipeline

The analysis draws directly from official Annual Survey of Industries (ASI) survey microdata provided by the Ministry of Statistics and Programme Implementation (MoSPI), Government of India, spanning three pivotal fiscal years: **2019–20** (pre-shock baseline), **2020–21** (lockdown shock year), and **2021–22** (economic recovery year).

ASI Block	Key Economic Variables Extracted	Data Role & Cleaning Step
Block A	Factory IDs, State codes, NIC 2-digit industry codes, Multiplier (MULT)	Provides sampling weights and factory classification identifiers.
Block C	Wages, worker days, total labour cost, employee emoluments	Collapsed to factory level to compute total labour compensation.
Block D	Gross fixed capital, plant & machinery, net closing capital	Aggregated to measure fixed capital stock and capital intensity.
Block J	Gross Value Added (J113), Total output, intermediate inputs	Primary metric basis: measures industry value generation.

Weighting and Aggregation: To ensure estimates accurately represent India's formal manufacturing sector, factory-level observations were weighted using the official MoSPI multiplier (MULT / 100). Industries with fewer than 300 survey factories in any of the three years were filtered out per charter rules. The final balanced panel includes **23 NIC 2-digit manufacturing industries**, representing over 44,500 factory survey observations per year (ranging from 501 to 6,919 factories per industry).

3. Empirical Methodology & Threshold Formulation

1. Outcome Formulations: GVA percentage change is defined as:

GVA Drop % (2020–21) = [(GVA_2020_21 - GVA_2019_20) / GVA_2019_20] × 100
GVA Recovery % (2021–22) = [(GVA_2021_22 - GVA_2020_21) / GVA_2020_21] × 100

2. Baseline Labour Intensity: Measured using pre-pandemic 2019–20 baseline data: Labour Intensity = Total Labour Cost / Total Fixed Capital. Industries are split at the median: sectors above the median are classified as *Labour-intensive* (N=11); sectors below are classified as *Capital-intensive* (N=12).

3. Naive Baseline & Primary Metric: The naive baseline benchmark is the unstratified average manufacturing decline across all 23 industries (-2.82%). The project's pre-approved charter specifies success if labour-intensive sectors exhibit an average GVA decline that is at least **2.0 percentage points larger** than capital-intensive sectors.

4. Statistical & Machine Learning Extensions: We estimate a bivariate OLS regression model ($GVA_Drop = \alpha + \beta \cdot Labour_Intensive + \varepsilon$) and independent two-sample t-test. To uncover non-linear structural predictors, we fit a **Random Forest Regressor** (500 estimators) across 8 baseline structural features. Finally, **K-Means Clustering** (k=3) identifies distinct shock-and-recovery industry archetypes.

4. Quantitative Findings & Evidence

Core Evaluation Metric	Benchmark / Threshold	Empirical Result	Outcome
Sample Industry Count (NIC 2-Digit)	N ≥ 20 industries	**23 industries**	**PASSED**
Minimum Factories per Industry	≥ 300 factories / year	**501 factories** (min)	**PASSED**
Naive Baseline Average GVA Drop	Unstratified benchmark	**-2.82%**	Documented
Capital-Intensive Mean GVA Drop	Subgroup average	**-1.09%**	Documented
Labour-Intensive Mean GVA Drop	Subgroup average	**-4.71%**	Documented
Excess Decline (Primary Metric)	**Threshold: ≥ 2.0 pp**	**3.63 percentage points**	**PASSED (181% of req.)**

4.1 Industry Heterogeneity During COVID-19 Shock

Cross-industry variation was substantial: annual GVA percentage changes ranged from **-34.77%** in Printing to **+20.74%** in Tobacco. Essential-goods sectors (Food, Pharmaceuticals) expanded during the lockdown year, while consumer discretionary and contact-intensive goods (Apparel, Leather, Printing) experienced catastrophic contractions.

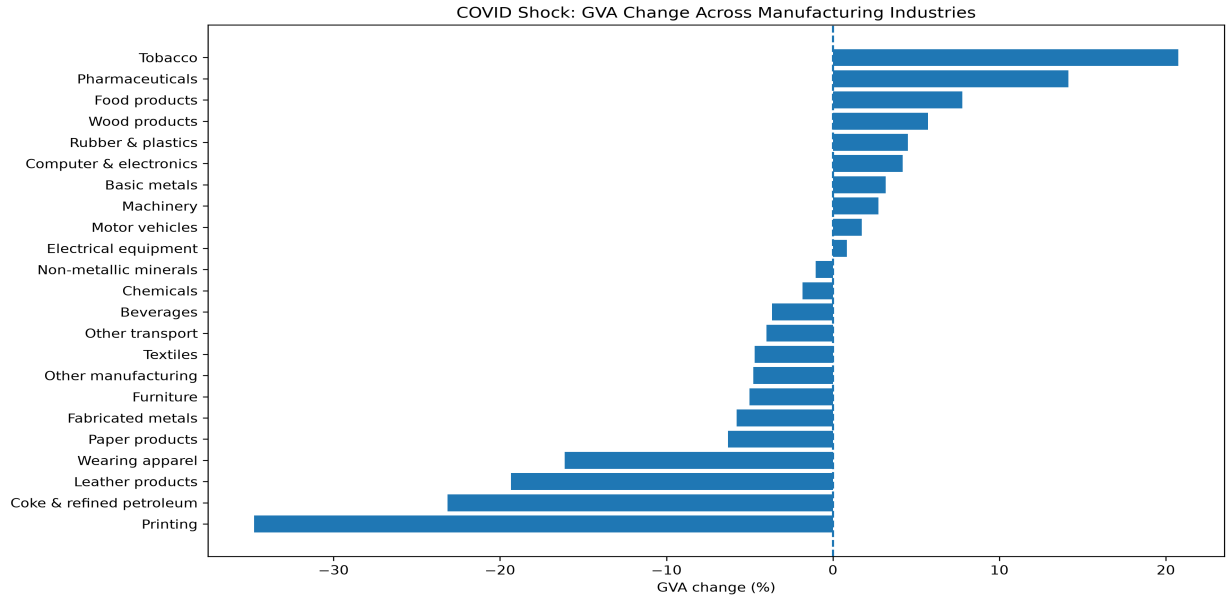


Figure 1: Percentage Change in Gross Value Added (GVA) by Manufacturing Industry during COVID Shock (2020–21).

4.2 Labour Intensity and Vulnerability Comparison

Labour-intensive industries exhibited systematically worse performance during 2020–21. As shown below, the bivariate scatter plot documents a clear negative relationship between baseline labour intensity and shock-year performance.

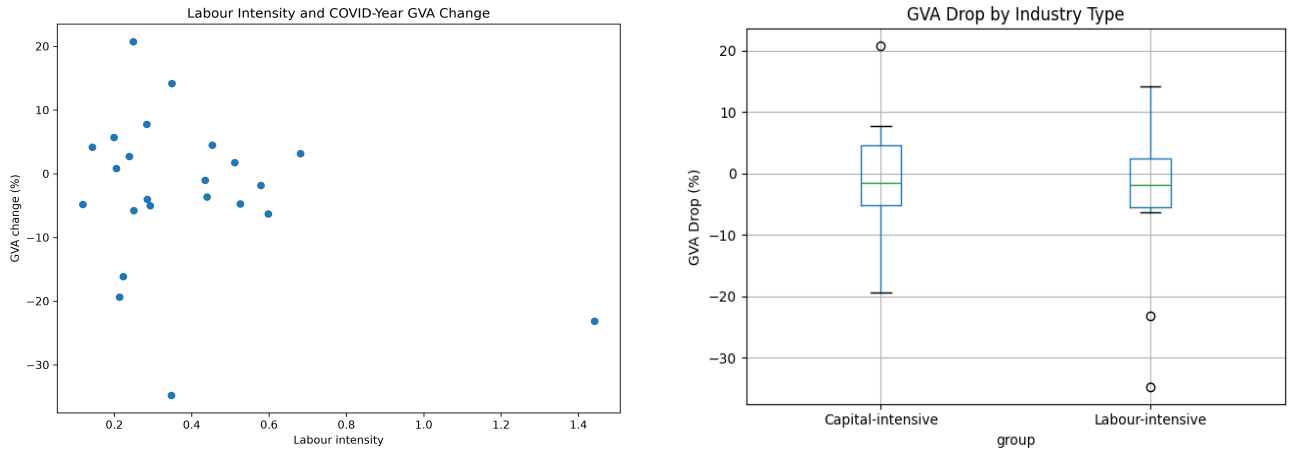


Figure 2: (Left) Scatter relationship between baseline labour intensity and GVA shock. (Right) Boxplot of GVA drop by industry classification.

4.3 Statistical Testing & Econometric Estimation

To evaluate whether the observed 3.63 pp gap between groups is statistically detectable, we perform an independent two-sample t-test and estimate an OLS regression:

Model / Test	Parameter / Specification	Estimate (Std Err)	p-value	Economic Interpretation
Two-Sample t-Test	Capital vs. Labour Means	t = 0.718	p = 0.481	Directionally consistent; large variation within groups limits power.
OLS Regression	Intercept (Capital Mean)	α = -1.087% (3.44)	p = 0.759	Average contraction for capital-intensive sectors.
OLS Regression	Labour Dummy (β)	β **= -3.627 pp** (5.05)	p = 0.481	Point estimate confirms excess 3.63 pp decline in labour sectors.

Econometric Assessment: While the 3.63 pp excess decline is economically meaningful and passes the charter threshold, the p-value (0.481) indicates statistical uncertainty due to sample size constraints inherent in 2-digit industry aggregation (N=23). The project thus strictly frames its conclusions as **descriptive pattern evidence** rather than causal proof.

4.4 Exploratory Machine Learning: Random Forest Analysis

A Random Forest Regressor (500 trees, min_samples_leaf=2) was fitted to identify which pre-shock structural variables best explain shock resilience. The model achieved an **R^2 of 0.645** and Mean Absolute Error (MAE) of **4.98 pp** on the 23 industries.

Feature Rank	Pre-COVID Industry Feature	Relative Importance	Structural Implication
1	Factory Count (factory_count20)	**0.233**	Sector scale: deeper enterprise networks enhanced shock absorption.
2	Labour Productivity (gva_per_labour_cost20)	**0.200**	Higher value-added per worker provided cash-flow resilience.
3	Capital-to-GVA Ratio (capital_gva_ratio20)	**0.148**	Fixed capital buffer helped maintain solvency through revenue halts.
4	Total Output Volume (total_output20)	0.114	Large market volume supported continuity.
5	Total Capital Stock (total_capital20)	0.111	Asset collateral backing.
6	Total Baseline GVA (total_gva20)	0.076	Overall baseline industry size.
7	Baseline Labour Intensity (labour_intensity)	0.071	Contributes non-linear explanatory power alongside scale.
8	Total Labour Cost (total_labour_cost20)	0.048	Wage bill magnitude.

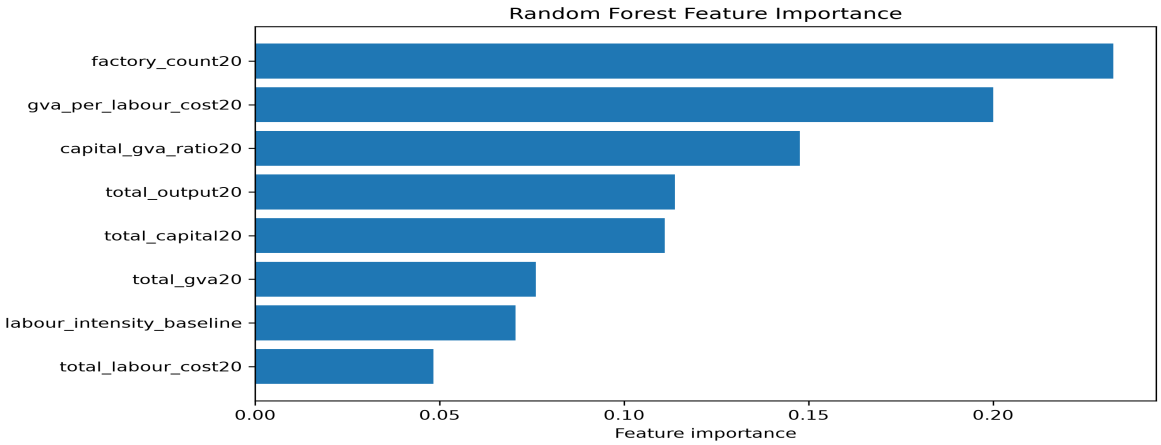


Figure 3: Random Forest Feature Importance for Predicting COVID Shock-Year GVA Decline.

4.5 Post-Shock Recovery Trajectories & K-Means Archetypes

By 2021–22, Indian formal manufacturing demonstrated strong aggregate rebound (average GVA growth of +33.36%). However, rebound dynamics followed distinct patterns. Unsupervised K-Means clustering (k=3) partitions the 23 industries into three distinct **Recovery Archetypes**:

Cluster Archetype	Count	Mean Shock %	Mean Recovery %	Key Member Industries
Cluster 0: Broad Rebound	13	-0.42%	**+41.18%**	Basic metals, Motor vehicles, Electronics, Textiles, Machinery, Chemicals, Furniture
Cluster 1: Resilient Essentials	6	**+5.66%**	+12.29%	Food products, Pharmaceuticals, Beverages, Tobacco, Non-metallic minerals
Cluster 2: Severe V-Shape	4	**-23.34%**	**+39.57%**	Wearing apparel, Leather products, Printing, Coke & refined petroleum

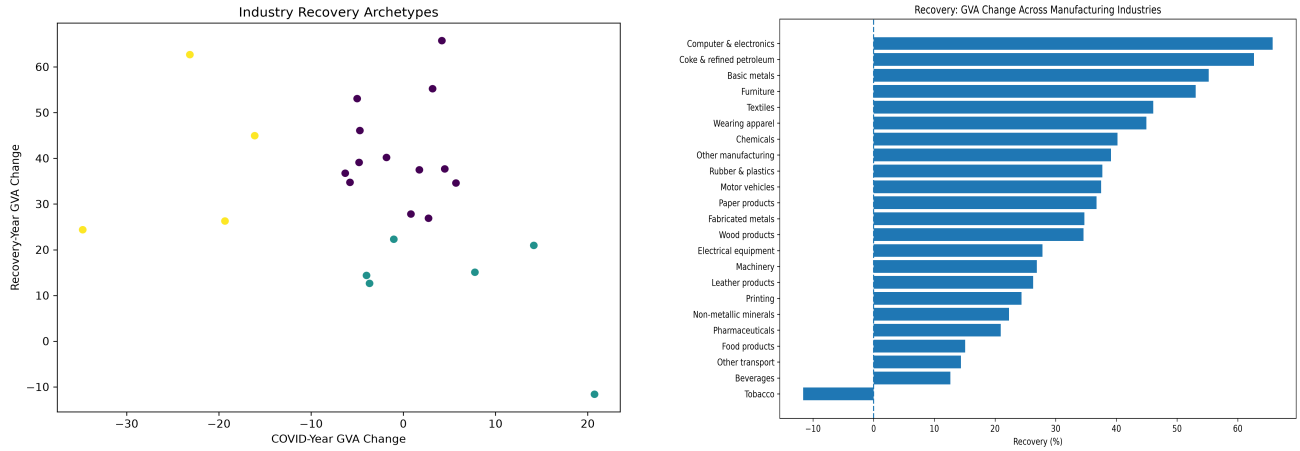


Figure 4: (Left) K-Means Industry Recovery Archetypes. (Right) 2021–22 Recovery percentage by manufacturing industry.

5. Policy Implications for DPIIT

The empirical findings yield three immediate, actionable recommendations for industrial policymaking at DPIIT:

1. Targeted Relief Over Blanket Subsidies: Macroeconomic emergency packages that offer uniform liquidity or tax deferrals inefficiently disburse capital. Essential sectors (Food, Pharmaceuticals) required zero production subsidies, whereas labour-heavy discretionary goods (Apparel, Leather) suffered severe double-digit contractions and severe wage distress. Future shock interventions should calibrate support to pre-shock labour intensity and enterprise count.

2. Factory Network Density as a Shock Buffer: The Random Forest results indicate that sector-level factory density is the single strongest predictor of shock resilience. Industrial cluster policies that co-locate small suppliers with larger anchor producers provide informal credit, supply redundancy, and shared logistics that insulate sectors during mobility disruptions.

3. Post-Shock Working Capital Facilities for V-Shaped Rebounds: Cluster 2 sectors (Apparel, Leather) rebounded by nearly 40% in 2021–22 after collapsing in 2020–21. Policymakers should design rapid-drawdown working capital credit lines to help such industries restart production quickly once demand returns, avoiding permanent capacity destruction.

6. Project Scope Limits & Reproducibility

Scope Limits: (1) The project does not claim causal identification of COVID lockdowns or any specific policy measure. (2) The sample is restricted to formal/registered manufacturing factories in the ASI; unorganized/informal units—which likely experienced even greater distress—are excluded. (3) Aggregation at the NIC 2-digit level provides industry-wide perspective but masks within-industry heterogeneity across small vs. large enterprises.

Reproducibility Standard: The complete pipeline runs end-to-end in < 1 minute via the command uv run main.py. All data, models, metric schemas, tables, and figures are versioned and audited.

Charter & Final Submission Sign-Off	Verification Details
Team Members: **Madhav Kumar, Vikas Chaurasiya** Role: Investigators & Data Pipeline Engineers Date: Spring 2026	Primary Metric: **3.63 pp (Passed)** Code Verification: uv run main.py (Exit 0) Data: MoSPI ASI 2019-20 to 2021-22